电势、电偶极矩

Faraday's law

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

若系统行为与时间无关，

$$\frac{\partial \vec{B}}{\partial t} = 0.$$

$$\nabla \times \vec{E} = 0.$$

静电场为无旋的.

另一方面，数学上，对任意标量场 φ，

$$\nabla \times \nabla \varphi = 0.$$

于是可引入标量场 φ，使得：

$$\vec{E} = -\nabla \varphi.$$

即，静电场可写为另一标量函数的负梯度.

定义电势 φ

$$\vec{E} \cdot d\vec{l} = -\left(\frac{\partial \varphi}{\partial x}\hat{e}_x + \frac{\partial \varphi}{\partial y}\hat{e}_y + \frac{\partial \varphi}{\partial z}\hat{e}_z\right) \cdot (dx \cdot \hat{e}_x + dy \cdot \hat{e}_y + dz \cdot \hat{e}_z)$$

$$= -\left(\frac{\partial \varphi}{\partial x}dx + \frac{\partial \varphi}{\partial y}dy + \frac{\partial \varphi}{\partial z}dz\right)$$

$$= -d\varphi.$$

对 $\forall$ 2点 P_1、P_2，

$$\varphi(P_1)-\varphi(P_2)=-\int_{P_2}^{P_1}\vec{E}\cdot d\vec{l}$$

积分与路径无关.

对有限分布电势体系，一般，我们取无穷远处电势为0. 即，

$$\varphi(\infty)=0.$$

$$\varphi(r)=\int_r^{\infty}\vec{E}\cdot d\vec{l}$$

$$\left.\begin{aligned}\nabla\cdot\vec{E}&=\frac{\rho}{\varepsilon_0}\\ \vec{E}&=-\nabla\varphi\end{aligned}\right\}\Rightarrow\quad \nabla\cdot\vec{E}=-\nabla^2\varphi=\frac{\rho}{\varepsilon_0}.$$

Poission eq

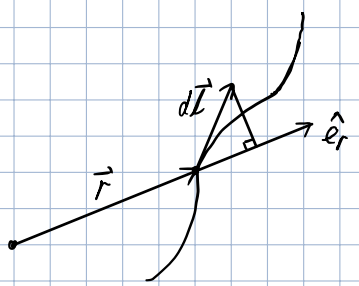
$$\nabla^2\varphi=-\frac{\rho}{\varepsilon_0}$$

讨论几种典型带电体系的电势.

1. 单个点电荷的电势.

$$\varphi(r)=\int_r^{\infty}\vec{E}\cdot d\vec{l}=\int_r^{\infty}\frac{q\hat{e}_r}{4\pi\varepsilon_0 r^2}\cdot d\vec{l}$$

$$=\int_r^{\infty}\frac{q}{4\pi\varepsilon_0 r^2}dr=\frac{q}{4\pi\varepsilon_0 r}$$

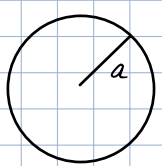


$\hat{e}_r\cdot d\vec{l}=dr$.

推广：若干点电荷组成带电体系.

$$\varphi=\sum_i\frac{q_i}{4\pi\varepsilon_0 r}$$

2. 半径为 a，电量 Q 均匀分布的球壳的内外电势.



① 球外.

$$\varphi(r)=\int_r^{\infty}\vec{E}\cdot d\vec{l}=\int_r^{\infty}\frac{Q}{4\pi\varepsilon_0 r^2}dr=\frac{Q}{4\pi\varepsilon_0 r}$$

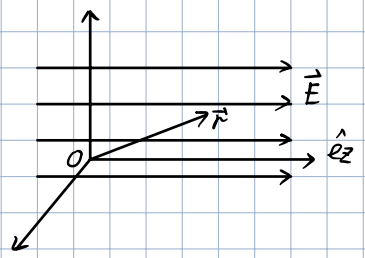
② 球内.

$$\varphi(r)=\int_r^{\infty}\vec{E}\cdot d\vec{l}=\int_a^{\infty}\vec{E}_1\cdot d\vec{l}+\int_r^{a}\vec{E}_2\cdot d\vec{l}$$

$$=\int_a^{\infty}\frac{Q\hat{e}_r}{4\pi\varepsilon_0 r}\cdot d\vec{l}+\int_r^{a}\frac{Q\vec{r}}{4\pi\varepsilon_0 a^3}\cdot d\vec{l}$$

$$=\frac{3Q}{8\pi\varepsilon_0 a}-\frac{Qr^2}{8\pi\varepsilon_0 a^3}$$

3. 均匀电场的电势.

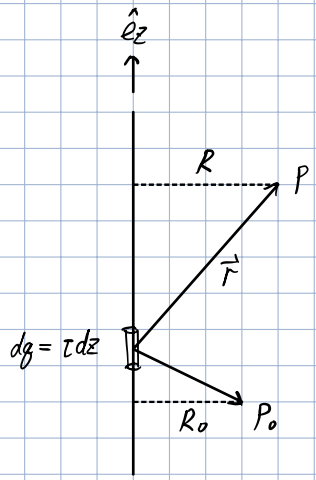


$$\vec{E} = E\hat{e}_z$$

$$\varphi(r) = \varphi(0) - \int_0^{\vec{r}} \vec{E}\cdot d\vec{l} = \varphi(0) - Ez$$

$$= -\vec{E}\cdot\vec{r}$$

4. 均匀带电的无限长直导线.



电荷线密度为 τ.

$$d\varphi = \frac{dq}{4\pi\varepsilon_0 r} = \frac{\tau dz}{4\pi\varepsilon_0\sqrt{R^2+z^2}}$$

$$\varphi(P) = \int_{-\infty}^{\infty} \frac{1}{4\pi\varepsilon_0}\frac{\tau}{\sqrt{R^2+z^2}}dz = \frac{\tau}{4\pi\varepsilon_0}\ln\left(z+\sqrt{R^2+z^2}\right)\Big|_{-\infty}^{\infty}.$$

上式结果为发散的，因为电荷并不分布于有限区域.

对策：取一点 P_0 令其电势为 0.

$$\varphi(P) - \varphi(P_0) = \frac{\tau}{4\pi\varepsilon_0}\left[\ln\left(z+\sqrt{R^2+z^2}\right)\Big|_{-\infty}^{\infty} - \ln\left(z+\sqrt{R_0^2+z^2}\right)\Big|_{-\infty}^{\infty}\right]$$

$$= \frac{\tau}{4\pi\varepsilon_0}\lim_{z\to\infty}\ln\frac{z+\sqrt{R^2+z^2}}{-z+\sqrt{R^2+z^2}}\cdot\frac{-z+\sqrt{R_0^2+z^2}}{z+\sqrt{R_0^2+z^2}}$$

$$= \frac{\tau}{4\pi\varepsilon_0}\lim_{z\to\infty}\ln\frac{1+\sqrt{(\frac{R}{z})^2+1}}{1+\sqrt{(\frac{R_0}{z})^2+1}}\cdot\frac{-1+\sqrt{(\frac{R}{z})^2+1}}{-1+\sqrt{(\frac{R_0}{z})^2+1}}.$$

利用 Taylor 展开.：

$$\sqrt{1+x^2} \approx 1+\frac{x^2}{2} \quad (x \ll 1)$$

$$\frac{1+\sqrt{(\frac{R}{z})^2+1}}{1+\sqrt{(\frac{R_0}{z})^2+1}}\cdot\frac{-1+\sqrt{(\frac{R}{z})^2+1}}{-1+\sqrt{(\frac{R_0}{z})^2+1}}\approx\frac{2+0.5(\frac{R}{z})^2}{2+0.5(\frac{R_0}{z})^2}\cdot\left(\frac{\frac{R_0}{z}}{\frac{R}{z}}\right)^2\Longrightarrow\left(\frac{R_0}{R}\right)^2$$

$$\varphi(P)=\frac{\tau}{2\pi\varepsilon_0}\ln\frac{R_0}{R}$$

由电势求场强：

柱坐标下，

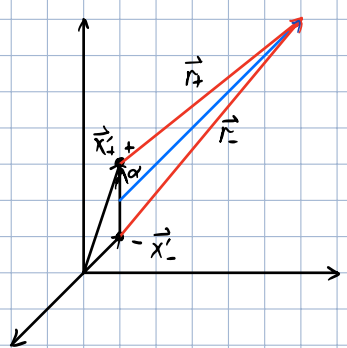
$$\nabla\varphi=\frac{\partial\varphi}{\partial R}\hat{e}_r+\frac{1}{R}\frac{\partial\varphi}{\partial\phi}\hat{e}_\phi+\frac{\partial\varphi}{\partial z}\hat{e}_z$$

$$\vec{E}_R=-\frac{\partial\varphi}{\partial R}=\frac{\tau}{2\pi\varepsilon_0 R}$$

$$E_\phi=E_z=0.$$

5. 电偶极子在远处的电势.

2 等量异号点电荷 q 与 $-q$ 构成一电偶极子.



定义 电偶极矩：

$$\vec{P}=\sum_{i=1}^{2}q_i\vec{x}_i'$$

$$\vec{l}=\vec{x}_+'-\vec{x}_-'$$

$$\vec{p}=\sum_i q_i\vec{x}_i=q\vec{x}_+'+(-q)\vec{x}_-'=q(\vec{x}_+'-\vec{x}_-')=q\vec{l}$$

$$\varphi=\frac{q}{4\pi\varepsilon_0}\frac{1}{r_+}+\frac{-q}{4\pi\varepsilon_0}\frac{1}{r_-}=\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_+}-\frac{1}{r_-}\right)$$

$$\approx\frac{q}{4\pi\varepsilon_0}\frac{l\cos\alpha}{R^2}=\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\vec{R}}{R^3}$$

$$\frac{1}{r_+}-\frac{1}{r_-}=\frac{r_--r_+}{r_+r_-}\approx\frac{r_--r_+}{R^2}$$

$$r_--r_+\approx l\cos\alpha.$$

$$\nabla(\vec{A}\cdot\vec{B}) = \vec{A}\times(\nabla\times\vec{B}) + (\vec{A}\cdot\nabla)\vec{B} + \vec{B}\times(\nabla\times\vec{A}) + (\vec{B}\cdot\nabla)\vec{A}$$

$$\vec{E} = -\nabla\varphi = -\frac{1}{4\pi\varepsilon_0}\nabla\left(\vec{p}\cdot\frac{\vec{R}}{R^3}\right)$$

$$= -\frac{1}{4\pi\varepsilon_0}\left[\vec{p}\times\left(\nabla\times\frac{\vec{R}}{R^3}\right) + (\vec{p}\cdot\nabla)\frac{\vec{R}}{R^3} + \frac{\vec{R}}{R^3}\times(\nabla\times\vec{p}) + \left(\frac{\vec{R}}{R^3}\cdot\nabla\right)\vec{p}\right]$$

$\left(\nabla\times\frac{\vec{R}}{R^3}\right) = 0$, $(\nabla\times\vec{p}) = 0$, $\left(\frac{\vec{R}}{R^3}\cdot\nabla\right)\vec{p} = 0$

$$= -\frac{1}{4\pi\varepsilon_0}\cdot\nabla\frac{\vec{R}}{R^3} = -\frac{\vec{p}}{4\pi\varepsilon_0}\cdot\left(R^{-3}\nabla\vec{R} + (\nabla R^{-3})\vec{R}\right)$$

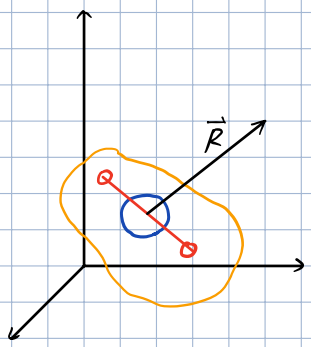
$$= -\frac{\vec{p}}{4\pi\varepsilon_0}\cdot\nabla\frac{\vec{R}}{R^3} = -\frac{\vec{p}}{4\pi\varepsilon_0}\cdot\left(R^{-3}\nabla\vec{R} + (\nabla R^{-3})\vec{R}\right)$$

$$= -\frac{\vec{p}}{4\pi\varepsilon_0}\cdot\left[-3\frac{\vec{R}\vec{R}}{R^5} + \frac{\overleftrightarrow{I}}{R^3}\right] = \frac{1}{4\pi\varepsilon_0}\left[\frac{3(\vec{p}\cdot\vec{R})}{R^5}\vec{R} - \frac{\vec{p}}{R^3}\right]$$

其中，$\nabla R^{-3} = \frac{dR^{-3}}{dR}\nabla R = -\frac{3}{R^4}\frac{\vec{R}}{R}$

$\nabla\vec{R} = \overleftrightarrow{I}$

任意形状带电体系的电势、电势的多极展开.



在无界空间中，Poission eq 的解就是

$$\varphi = \sum_i \frac{q_i}{4\pi\varepsilon_0 r_i} = \int \frac{\rho(\vec{x}')}{4\pi\varepsilon_0 r} dV'$$

一般情况下，系统的形状不具对称性，积分是困难的.

① 最简单、粗糙的近似，将其视为点电荷.

$$\varphi^{(0)} = \frac{Q}{4\pi\varepsilon_0 R} \begin{cases} Q = \sum_i q_i & \text{分立电荷分布系统} \\ Q = \int \rho(\vec{x}') dV' & \text{连续电荷分布系统.} \end{cases}$$

② 进一步地，我们将带电体视为电偶极子.

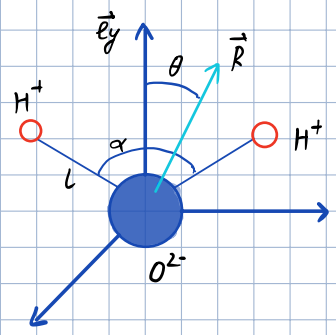
$$\varphi^{(1)} = \frac{1}{4\pi\varepsilon_0} \frac{\vec{p}\cdot\vec{R}}{R^3} \begin{cases} \vec{p} = \sum_{i=1}^{n} q_i \vec{x}_i & \text{离散} \\ \vec{p} = \int \rho(\vec{x}')\vec{x}' dV'. & \text{连续.} \end{cases}$$

③ 更高级的修正，将带电体看成一电四极矩.

$$\varphi = \varphi^{(0)} + \varphi^{(1)} + \varphi^{(2)} + \cdots$$

$$\varphi^{(0)} \sim \frac{1}{R} \quad \varphi^{(1)} \sim \frac{1}{R^2} \quad \varphi^{(2)} \sim \frac{1}{R^3}$$

例：水分子的远场电势.



$$Q = -2e + 2\times e = 0$$

$$\varphi^{(0)} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{R} = 0$$

$$\vec{P} = \sum_i q_i \vec{r_i'} = -2e\times 0 + e\vec{r_1} + e\vec{r_2}$$

$$= e(\vec{r_1} + \vec{r_2}) = 2el\cdot\cos\frac{\alpha}{2}\cdot\hat{e_y}$$

$$\varphi \approx \varphi^{(1)} = \frac{1}{4\pi\varepsilon_0}\frac{\vec{P}\cdot\vec{R}}{R^3} = \frac{el\cos\frac{\alpha}{2}\cos\theta}{2\pi\varepsilon_0 R^2}$$

电势的多极展开.

考虑单变量函数 $f(x-x')$

在 x 附近 $(x'=0)$ 作 Taylor 展开.

$$f(x-x') = f(x) + \left.\frac{\partial f}{\partial x}\right|_{x'=0}(x-x'-x) + \frac{1}{2}\left.\frac{\partial^2 f}{\partial x^2}\right|_{x'=0}(x'x') + \cdots$$

$$= f(x) - \left.\frac{\partial f}{\partial x}\right|_{x'=0}\cdot x' + \frac{1}{2}\left.\frac{\partial^2 f}{\partial x^2}\right|_{x'=0}(x'x') + \cdots$$

推广至三维空间：

$$f(\vec{x}-\vec{x'}) = f(\vec{x}) - \sum_{i=1}^{3}\left.\frac{\partial f}{\partial x_i}\right|_{x'=0}\cdot x_i' + \frac{1}{2}\sum_{ij}\left.\frac{\partial^2 f}{\partial x_i\partial x_j}\right|_{x'=0}\cdot(x_i'x_j') + \cdots$$

$$= f(\vec{x}) - x_i'\cdot\nabla f\big|_{x'=0} + \frac{1}{2}\sum_{ij}\left.\frac{\partial^2 f}{\partial x_i\partial x_j}\right|_{x'=0}\cdot(x_i'x_j') + \cdots$$

令 $f(\vec{x}-\vec{x}') = \frac{1}{r}$

$$\frac{1}{r} = \frac{1}{R} - \vec{x}'\cdot\nabla\frac{1}{R} + \frac{1}{2}\sum_{ij} x_i' x_j' \frac{\partial^2 \frac{1}{R}}{\partial x_i \partial x_j} + \cdots$$

$$= \frac{1}{R} + \frac{\vec{x}'\cdot\vec{R}}{R^3} + \frac{1}{2}\sum_{ij} x_i' x_j' \frac{\partial^2 \frac{1}{R}}{\partial x_i \partial x_j} + \cdots$$

$$\varphi = \int \frac{\rho(\vec{x}')}{4\pi\varepsilon_0}\left[\frac{1}{R} + \frac{\vec{x}'\cdot\vec{R}}{R^3} + \frac{1}{2}\sum_{ij} x_i' x_j' \frac{\partial^2 \frac{1}{R}}{\partial x_i \partial x_j} + \cdots\right] dV'$$

$$= \int \frac{\rho(\vec{x}')}{4\pi\varepsilon_0}\left[\frac{1}{R} + \frac{\vec{x}'\cdot\vec{R}}{R^3} + \frac{1}{2}\sum_{ij} x_i' x_j' \frac{\partial^2 \frac{1}{R}}{\partial x_i \partial x_j} + \cdots\right] dV'$$

$$= \int \frac{\rho(\vec{x}')}{4\pi\varepsilon_0 R} dV' + \int \frac{\rho}{4\pi\varepsilon_0}\frac{\vec{x}'\cdot\vec{R}}{R^3} dV' + \int \frac{1}{8\pi\varepsilon_0}\left[\sum_{ij} x_i' x_j' \rho(\vec{x}')\right] dV' \frac{\partial^2 \frac{1}{R}}{\partial x_i \partial x_j} + \cdots$$

$$= \frac{Q}{4\pi\varepsilon_0 R} + \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\vec{R}}{R^3} + \frac{1}{24\pi\varepsilon_0}\overleftrightarrow{D} : \nabla\nabla\frac{1}{R} + \cdots$$

$$= \varphi^{(0)} + \varphi^{(1)} + \varphi^{(2)} + \cdots$$

- 电偶极矩 p 是衡量系统电荷偏离中心对称分布的一个物理量。对具球对称分布的带电系统，$p=0$.
- 电四极矩 $\overleftrightarrow{D}$ 是衡量系统电荷偏离球对称分布的一个物理量.

$\nabla\nabla\frac{1}{R}$ 为张量、 $\overleftrightarrow{D} : \nabla\nabla\frac{1}{R}$ 为标量.

关于电荷系统的电四极矩：

$$\overleftrightarrow{D} = \int 3\rho(\vec{x}')\,\vec{x}'\vec{x}'\,dV'$$

（$\overleftrightarrow{D}$：张量；$\vec{x}'\vec{x}'$：并矢）

分量形式：$D_{ij} = \int 3\rho(\vec{x}')\,x_i' x_j'\,dV'$

对于分立电荷系统

$$\overleftrightarrow{D} = \sum_k 3q_k \vec{x}_k'\vec{x}_k'$$

$$D_{ij} = \sum_k 3q_k (x_i' x_j')_k$$

电四极矩为对称张量，$D_{ij} = D_{ji}$，有6个独立分量：

$$D_{xx},\ D_{yy},\ D_{zz},\ D_{xy},\ D_{xz},\ D_{yz}.$$

注意到，

$$0 = \nabla^2 \frac{1}{R} = \nabla\cdot\nabla\frac{1}{R} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\frac{1}{R} = \sum_i \frac{\partial^2}{\partial x_i^2}\frac{1}{R}$$

$$0 = r'^2\nabla^2\frac{1}{R} = r'^2\sum_i \frac{\partial^2}{\partial x_i^2}\frac{1}{R} = r'^2\sum_{ij}\delta_{ij}\frac{\partial^2}{\partial x_i\partial x_j}\frac{1}{R}.$$

6个独立分量可消去1个.

$$\varphi^{(2)} = \frac{1}{24\pi\varepsilon_0}\int\left[\sum_{ij} 3x_i' x_j'\rho(\vec{x}')\frac{\partial\frac{1}{R}}{\partial x_i\partial x_j}\right]dV'$$

$$\Rightarrow \quad \varphi^{(2)} = \frac{1}{24\pi\varepsilon_0}\sum_{ij}\left[\int(3x_i'x_j' - r'^2\delta_{ij})\rho(\vec{x}')\,dV'\right]\frac{\partial\frac{1}{R}}{\partial x_i\partial x_j}$$

由此，重新定义电四极矩.

$$D_{ij} = \int(3x_i'x_j' - r'^2\delta_{ij})\rho(\vec{x}')\,dV'.$$

其中，$r'^2 = x'^2 + y'^2 + z'^2$

$$\varphi^{(2)} = \frac{1}{24\pi\varepsilon_0} \sum_{ij} D_{ij} \frac{\partial \frac{1}{R}}{\partial x_i \partial x_j}$$

$$= \frac{1}{24\pi\varepsilon_0 R^5} \left[(3x^2 - R^2) D_{xx} + (3y^2 - R^2) D_{yy} + (3z^2 - R^2) D_{zz} + 6xy D_{xy} + 6xz D_{xz} + 6yz D_{yz} \right]$$

静电场边值问题.

问题：① 空间中仅一种介质.

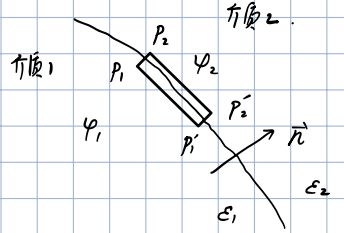
$$\rho(\vec{x}') \longrightarrow \varphi = \int \frac{\rho(\vec{x}')}{4\pi\varepsilon_0 r} dV' \longrightarrow \vec{E} = -\nabla\varphi.$$

② 存在不同介质？

介质分界面或内部有极化电荷分布存在.

这些电荷不能先于电场求出，要研究在一定边界条件下场或势微分方程的解.

场方程
$$\begin{cases} \nabla\cdot\vec{D} = \rho_f \\ \nabla\times\vec{E} = -\dfrac{\partial\vec{B}}{\partial t} \end{cases}$$



边值关系
$$\begin{cases} \vec{n}\cdot(\vec{D}_2 - \vec{D}_1) = \sigma_f \\ \vec{n}\times(\vec{E}_2 - \vec{E}_1) = 0 \end{cases}$$

$$E = -\nabla\varphi.$$

各向同性线性理想介质中，

$$\vec{D} = \varepsilon\vec{E}.$$

$$\nabla\cdot\vec{D} = \nabla\cdot\varepsilon\vec{E} = \varepsilon\nabla\cdot\vec{E} = -\varepsilon\nabla^2\cdot\varphi = \rho_f$$

于是有：

各向同性线性理想介质中的 Poission eq.

$$\nabla^2\varphi = -\frac{\rho_f}{\varepsilon}$$

在 $\rho_f = 0$ 区域，有

Laplace's eq

$$\nabla^2\varphi = 0$$

边界条件：

$$\vec{n}\cdot(\vec{D}_2 - \vec{D}_1) = \vec{n}\cdot(\varepsilon_2\vec{E}_2 - \varepsilon_1\vec{E}_1) = \vec{n}\cdot(-\varepsilon_2\nabla\varphi_2 + \varepsilon_1\nabla\varphi_1) = \sigma_f$$

$\Longleftrightarrow$

$$-\varepsilon_2\frac{\partial\varphi_2}{\partial n} + \varepsilon_1\frac{\partial\varphi_1}{\partial n} = \sigma_f$$

在边界 2 侧取无限薄环路：

$$\overline{P_1P_2} = \overline{P_1'P_2'} \to 0. \quad \overline{P_1P_1'} = \overline{P_2P_2'} = \Delta l.$$

$$d\varphi = -\vec{E}\cdot d\vec{l}$$

$$\varphi_2(P_2) - \varphi_1(P_1) = \vec{E}\cdot\overline{P_1P_2} \to 0$$

∴ 有：

$$\begin{cases} \varphi_2(P_2) = \varphi_1(P_1) \\ \varphi_2'(P_2') = \varphi_1'(P_1') \end{cases}$$

（电势的连续性）.

于是，

$$\varphi_2 - \varphi_1 = \varphi_2' - \varphi_1'$$

$$E_{2\tau} \cdot \Delta l = E_{1\tau} \cdot \Delta l$$

$$\vec{E}_2 \cdot \Delta \vec{l} = \vec{E}_1 \cdot \Delta \vec{l}$$

$$E_{2\tau} = E_{1\tau}$$

亦即 边值关系 $\varphi_2 = \varphi_1 \iff n \times (\vec{E}_2 - \vec{E}_1) = 0$

总结

电场方程 ⟺ 电势方程

$$\begin{cases} \nabla \cdot \vec{D} = \rho_f \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \end{cases} \iff \nabla^2 \varphi = -\frac{\rho_f}{\varepsilon}$$

边界条件

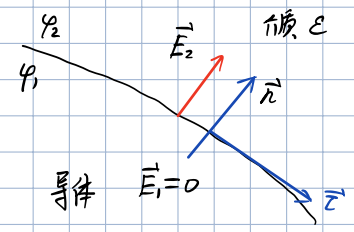
$$\begin{cases} n \cdot (\vec{D}_2 - \vec{D}_1) = \sigma_f \\ n \times (\vec{E}_2 - \vec{E}_1) = 0 \end{cases} \iff \begin{cases} -\varepsilon_2 \dfrac{\partial \varphi_2}{\partial n} + \varepsilon_1 \dfrac{\partial \varphi_1}{\partial n} = \sigma_f \\ \varphi_2 = \varphi_1 \end{cases}$$

有导体存在时情形

导体静电平衡条件：

(1) 导体内部电场 E 处处为 0. 外侧只有 $\vec{E}$ 和 $\vec{D}$ 的法向分量.

(2) 导体内部自由电荷密度 ρ_f 处 ξ 为 0, 电荷仅以面密度 σ_f 分布于表面.



导体为一等势体

$$\varphi_1 = Const$$

$$\vec{E}_1 = -\nabla\varphi_1 = 0$$

由电势连续性，导体表面电势 $\equiv$ 内部电势.

$$\varphi_{inside} = \varphi_{surface} = const$$

另一边界条件

$$-\varepsilon_2 \frac{\partial\varphi_2}{\partial n} = \sigma_f$$

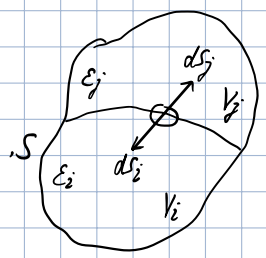
总结

导体表面边值关系

$$\varphi_1 = const$$

$$-\varepsilon_2 \frac{\partial\varphi_2}{\partial n} = \sigma_f$$

静电边值问题解的唯一性定理.



设介质分区均匀，区域 V_i 内电荷分布确定，有：

$$\nabla^2\varphi_i = -\frac{\rho_f}{\varepsilon_i}$$

在某2个均匀区域 V_i 与 V_j 的分界面上，有边值关系

$$\varphi_1 = \text{const}$$

$$-\varepsilon_2\frac{\partial\varphi_2}{\partial n} = \sigma_f$$

除了上面两式，要确定 V 内电场，还需给出 V 的边界 S 上的条件.

唯一性定理 指出：

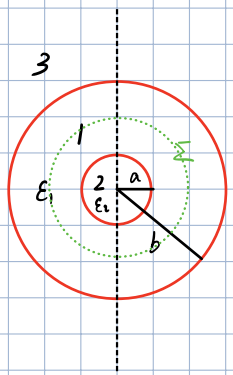
设区域 V 内给定自由电荷分布 $\rho(x)$，在 V 的边界 S 上只要给定：

(1) 电势 $\varphi|_S$

(2) 电势的法线方向偏导数 $\frac{\partial\varphi}{\partial n}\Big|_S$.

V 内的电场即可唯一确定

例：同心导体球，内球壳带电Q，外球壳接地，求电场与球壳电荷分布.



解：区域3：$\vec{E}_3 = 0 \quad \vec{D}_3 = 0 \quad \varphi_3 = 0$

介质1：$\nabla \times \vec{E}_1 = 0 \quad \nabla \cdot \vec{D}_1 = 0$

介质2：$\nabla \times \vec{E}_2 = 0 \quad \nabla \cdot \vec{D}_2 = 0$

介质1,2分界面

$$E_{1r} = E_{2r}$$

$$D_{2n} - D_{1n} = \sigma_{f} = 0$$

不解方程，用唯一性定理猜出这个解：

$$\vec{E}_1 = \frac{A}{r^2}\hat{e}_r = \vec{E}_2$$

$$\oint_{\Sigma} \vec{D} \cdot d\vec{\Sigma} = \iint_1 \varepsilon_1 \vec{E}_1 \cdot d\vec{\Sigma}_1 + \iint_2 \varepsilon_2 \vec{E}_2 \cdot d\vec{\Sigma}_2$$

$$= (\varepsilon_1 E_1 + \varepsilon_2 E_2)\, 2\pi r^2 = Q.$$

$$\Longrightarrow \quad A = \frac{Q}{2\pi(\varepsilon_1 + \varepsilon_2)}$$

解得：

$$\vec{E}_1 = \vec{E}_2 = \frac{Q}{2\pi(\varepsilon_1 + \varepsilon_2) r^2}\hat{e}_r$$

$$\vec{D}_1 = \frac{\varepsilon_1 Q}{2\pi(\varepsilon_1 + \varepsilon_2) r^2}\hat{e}_r$$

$$\vec{D}_2 = \frac{\varepsilon_2 Q}{2\pi(\varepsilon_1 + \varepsilon_2)}\hat{e}_r$$

唯一性定理保证，该解必然正确.

边界问题解法1. 分离变量法.

若求解区域 V 内，电荷体密度 $\rho = 0$

该区域电势满足 Laplace eq

$$\nabla^2 \psi = 0$$

在有轴对称情况下，Laplace eq 在球坐标下写为：

$$\frac{\partial}{\partial r}\left(r^2 \frac{\partial \psi}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \psi}{\partial \theta}\right) = 0$$

分离变量法：设 $\psi(r,\theta) = R(r)\,\Theta(\theta)$

$$\frac{1}{R}\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) = -\frac{1}{\Theta \sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right)$$

左边只与 r 有关，右边只与 θ 有关. 故两式恒相等于常数.

令其恒等于 $n(n+1)$

$$\frac{d}{dr}\left(r^2 \frac{dR}{dr}\right) - n(n+1)R = 0$$

$$\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + n(n+1)\sin\theta\,\Theta = 0$$

第1个方程易求解为：

$$R = a_n r^n + \frac{b_n}{r^{n+1}}$$

其中，a_n、b_n 为常数.

对第2个方程，作变量替换. $\zeta = \cos\theta$

$$\frac{d}{d\zeta}\left[(1-\zeta^2)\frac{d\Theta}{d\zeta}\right] + n(n+1)\Theta = 0$$

此方程为 勒让德方程. 只有当 n 为整数才存在 $-1 \le \zeta \le 1$ 区间内的有限解.

其解为勒让德多项式，记为

$$\Theta = P_n(\cos\theta)$$

故方程通解为：

$$\psi_n(r,\theta) = \sum_{n=0}^{\infty}\left(a_n r^n + \frac{b_n}{r^{n+1}}\right) P_n(\cos\theta)$$

求解 $P_n(\cos\theta)$ 的显式：

当 $r\neq 0$ 时，单位点电荷电势 $\frac{1}{r}$ 为 Laplace eq 的解.

对 Laplace eq 作用算符 $\frac{\partial}{\partial z}$.

$$\frac{\partial}{\partial z}\nabla^2\psi = \nabla^2\frac{\partial}{\partial z}\psi = 0$$

∴ 若 ψ 为方程的解，$\frac{\partial\psi}{\partial z}$ 也为方程的解.

同理，$\frac{\partial^n\psi}{\partial z^n}$ 也为方程的解.

∴ Laplace eq 具有特解

$$\frac{1}{r}$$

$$\frac{\partial}{\partial z}\left(\frac{1}{r}\right) = -\frac{z}{r^3} = -\frac{1}{r^2}\cos\theta$$

$$\frac{\partial^2}{\partial z^2}\left(\frac{1}{r}\right) = \frac{3z^2-r^2}{r^5} = \frac{1}{r^3}(3\cos^2\theta - 1)$$

……

上述特解均具备形式：

$$\frac{1}{r^{n+1}}P_n(\cos\theta).$$

对比即有：

$$P_0(\cos\theta) = 1$$

$$P_1(\cos\theta) = \cos\theta$$

$$P_2(\cos\theta) = \frac{1}{2}(3\cos^2\theta - 1)$$

$$P_3(\cos\theta) = \frac{1}{2}(5\cos^3\theta - 3\cos\theta)$$

于是一般表达式：

$$P_n(\cos\theta) = \frac{1}{2^n \cdot n!} \frac{d^n}{d(\cos\theta)^n}[(\cos^2\theta - 1)^n]$$

镜像法

镜像法：

用一个或多个假想的点电荷来代替（等效）导体表面感应电荷或介质的极化电荷对电场的贡献.

只要这些假想电荷与原来已知电荷共同激发的电场或电势满足求解区域内的全部（定解）边界条件，那么所得到的解是唯一正确的.

- 用像电荷代替感应电荷，像电荷与原电荷势点电荷，电势容易计算.
- 为使问题的解满足求解区域内已知的 Poisson 方程或 Laplace 方程，像电荷必须放置于求解区域外.

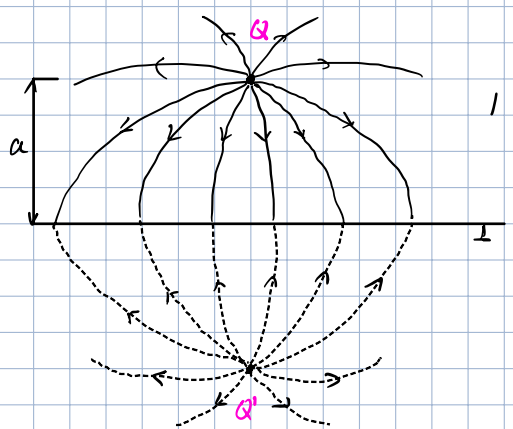
例一：接地无限大平面导体板附近有一点电荷 Q. 求空间中电场.

导体表面为等势面.

求解区域为上半空间.

区域 1：$\nabla^2\varphi = 0$ （除 Q）

区域 2：$\varphi \equiv 0$.



由边界条件，导体内外 $\vec{E}_t$ 连续，但导体内部无电场，故外面 $\vec{E}_t = 0$

边界条件：

1) 电力线 ⊥ 导体表面.

2) $\varphi|_{R\to\infty} = 0$.

镜像法：

在下半平面对称处放置一假想点电荷 $-Q$.

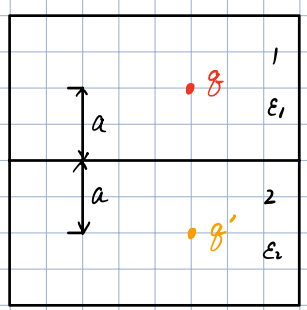
用假想电荷代替分界面上真实的感应电荷.

于是上半空间电势

$$\varphi(P) = \frac{q}{4\pi\varepsilon_0 r} - \frac{q}{4\pi\varepsilon_0 r'}$$

推广：

上、下半空间分别充满了介电常数分别为：ε_1，ε_2 的均匀介质，在 $z=a$ 处有一点电荷. 求电荷分布，以及电荷所受作用力.



解：　方程

$$\nabla^2\varphi_{1,2} = 0 \quad (q\text{点除外})$$

边界条件：

$$\varphi_1|_{z=0} = \varphi_2|_{z=0}$$

$$\varepsilon_1\frac{\partial\varphi_1}{\partial z}\Big|_{z=0} = \varepsilon_2\frac{\partial\varphi_2}{\partial z}\Big|_{z=0}.$$

对区域 1，在区域 2：$z=-a$ 处放置一像电荷 q'

$$\varphi_1 = \frac{q}{4\pi\varepsilon_1\sqrt{x^2+y^2+(z-a)^2}} + \frac{q'}{4\pi\varepsilon_1\sqrt{x^2+y^2+(x+a)^2}}$$

对区域2，在区域1中 $z=a$ 原电荷 q 处再放置一像电荷 q''.

$$\varphi_2 = \frac{q+q''}{4\pi\varepsilon_2\sqrt{x^2+y^2+(z-a)^2}}$$

根据边界条件：

$$\varphi_1|_{z=0} = \varphi_2|_{z=0}$$

$$\Rightarrow \quad \frac{q}{4\pi\varepsilon_1\sqrt{x^2+y^2+a^2}} + \frac{q'}{4\pi\varepsilon_1\sqrt{x^2+y^2+a^2}} = \frac{q+q''}{4\pi\varepsilon_2\sqrt{x^2+y^2+a^2}}$$

$$\varepsilon_1\frac{\partial\varphi_1}{\partial z}|_{z=0} = \varepsilon_2\frac{\partial\varphi_2}{\partial z}|_{z=0}$$

$$\Rightarrow \quad \frac{-aq}{(x^2+y^2+a^2)^{\frac{3}{2}}} + \frac{aq'}{(x^2+y^2+a^2)^{\frac{3}{2}}} = \frac{-a(q+q'')}{(x^2+y^2+a^2)^{\frac{3}{2}}}$$

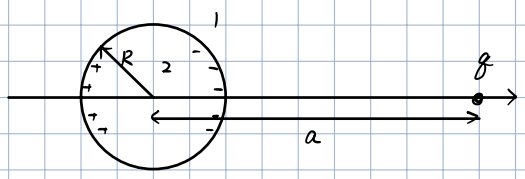
解得：

$$q' = \frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+\varepsilon_2}q \qquad q'' = -q'$$

电荷 q 受力

$$\vec{F} = q\vec{E}_{q'} = \frac{qq'\hat{e}_z}{4\pi\varepsilon_0(2a)^2} = \frac{q^2(\varepsilon_1-\varepsilon_2)}{16\pi\varepsilon_1 a^2(\varepsilon_1+\varepsilon_2)}\hat{e}_z$$

例 2　半径为 R_0 的接地导体球外有一点电荷 q，求电势



解：导体球为等势体，求解区域为球外空间.

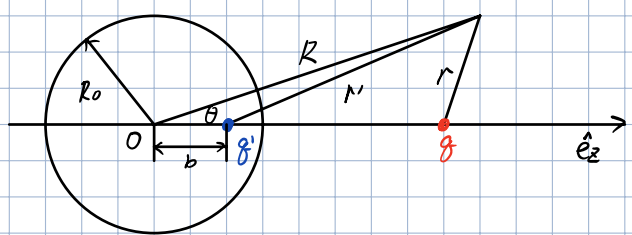
区域 1.　满足 $\nabla^2\varphi = 0$（除 q 同）

区域 2.　电势恒为 0.

球外电场 = q 贡献电场 + 导体球面出现感应电荷的场.

感应电荷 $\Longleftrightarrow$ 假想的像电荷 q' 产生的场.

由对称性，q' 应放在 z 轴.



边界条件：

$$\varphi|_{R_0} = 0$$

$$\Longrightarrow \quad \frac{q}{4\pi\varepsilon_0 r}\Big|_{R_0} + \frac{q'}{4\pi\varepsilon_0 r'} = 0 \quad \Longrightarrow \quad \frac{q^2}{r^2}\Big|_{R_0} = \frac{q'^2}{r'^2}\Big|_{R_0}$$

代入几何关系有：　$b = \frac{R_0^2}{a}$　　$q' = -\frac{R_0}{a}q$

$$\varphi = \frac{q}{4\pi\varepsilon_0\sqrt{R^2+a^2-2Ra\cos\theta}} + \frac{-\frac{R_0 q}{a}}{4\pi\varepsilon_0\sqrt{R^2+(\frac{R_0^2}{a})^2-\frac{2R_0R}{a}\cos\theta}}$$

感应电荷面密度.

$$\sigma_{\text{感}} = \vec{n}\cdot(\vec{D}_2-\vec{D}_1) = \vec{D}_2\cdot\hat{e}_r = \varepsilon_0\nabla\varphi = -\varepsilon_0\frac{\partial\varphi}{\partial R}\Big|_R$$

例3 若导体球不接地？

(1) 此时 $\varphi|_{R_0} = \text{const}$.

除在 b 处之外，再在球心处放置一点电荷

同上　　另计算.

(2) 边界条件还要

$$\oint_{\text{surface}} \vec{D}\cdot d\vec{\Sigma} = -\oint_{\text{surface}} \varepsilon_0 \frac{\partial \varphi}{\partial R}\, dS = q_0$$

Green 函数解

若非齐次偏微分方程非齐次项为 δ 函数，满足边界条件的方程的定解称 Green 函数解.

例如：

在 $\vec{x}'$ 处有点电荷，电势满足 Poisson 方程.

$$\nabla^2\varphi = -\frac{1}{\varepsilon}\delta(\vec{x}-\vec{x}')$$

且满足边界条件：

$$\varphi|_S = 0 \quad 或 \quad \frac{\partial\varphi}{\partial n}\Big|_S = 0$$

则 φ 为 Green function，$\varphi = G(\vec{x}, \vec{x}')$

即：

$$\begin{cases} \nabla^2 G(\vec{x}-\vec{x}') = -\delta(\vec{x}-\vec{x}')/\varepsilon \\ G|_S = 0 \quad 或 \quad \frac{\partial G}{\partial n}\Big|_S = 0 \end{cases}$$

第一类边界问题　　第二类边界问题

例：无界空间的 Green 函数

$$G(\vec{x}, \vec{x}') = G(r) = \frac{1}{4\pi\varepsilon_0 r}$$

$r \neq 0$，$\nabla^2\frac{1}{r} = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\right] = 0$

$r = 0$，$\lim\limits_{V\to 0}\int_V \nabla^2\frac{1}{r}dV = \int_V \nabla\cdot\nabla\frac{1}{r}dV = \oint\nabla\frac{1}{r}\cdot d\vec{S} = -\oint\frac{\hat{e}_r}{r^2}dS = -\oint\frac{1}{r^2}\cdot r^2 d\Omega = -4\pi.$

$$\nabla^2\frac{1}{r} = -4\pi\delta(r)$$

$$\nabla^2 G(r) = -\frac{\delta(r)}{\varepsilon_0}$$

利用上面求边值问题的解.

设 $\varphi(\vec{x})$, $\psi(\vec{x})$ 满足.

Poisson eq $\quad \nabla^2\varphi = -\frac{\rho(\vec{x}')}{\varepsilon_0}$

无界空间 Green function $\quad \nabla^2\psi = -\frac{\delta(\vec{x}-\vec{x}')}{\varepsilon_0}$

考虑:

① $\quad \nabla\cdot(\psi\nabla\varphi) = \nabla\psi\cdot\nabla\varphi + \psi\nabla^2\varphi$

② $\quad \nabla\cdot(\varphi\nabla\psi) = \nabla\psi\cdot\nabla\varphi + \varphi\nabla^2\psi$.

①-② $\quad \nabla\cdot(\psi\nabla\varphi - \varphi\nabla\psi) = \psi\nabla^2\varphi - \varphi\nabla^2\psi$.

积分:

$$\iiint_V(\psi\nabla^2\varphi - \varphi\nabla^2\psi)dV' = \iiint_V\nabla\cdot(\psi\nabla\varphi - \varphi\nabla\psi)dV' \overset{\text{Gauss's law}}{=\!=} \oiint(\psi\nabla\varphi - \varphi\nabla\psi)\cdot d\vec{S}'$$

将 φ, ψ 代入, 有:

$$\iiint_V\left(-G(\vec{x},\vec{x}')\frac{\rho(\vec{x}')}{\varepsilon_0} - \varphi(\vec{x}')\nabla^2G(\vec{x},\vec{x}')\right)dV' = \oiint_S\left(G(\vec{x},\vec{x}')\frac{\partial\varphi}{\partial n} - \varphi\frac{\partial G}{\partial n}\right)\cdot d\vec{S}'$$

其中,

$$\iiint_V\left(-\varphi(\vec{x}')\nabla^2G(\vec{x},\vec{x}')\right)dV' = \iiint_V\varphi(\vec{x}')\frac{\delta(\vec{x}-\vec{x}')}{\varepsilon_0}dV' = \frac{\varphi(\vec{x})}{\varepsilon_0}$$

于是有

格林公式:

$$\varphi(\vec{x}) = \iiint_V G(\vec{x},\vec{x}')\rho(\vec{x}')dV' + \varepsilon_0\oiint\left(G(\vec{x},\vec{x}')\frac{\partial\varphi}{\partial n} - \varphi(\vec{x}')\frac{\partial G(\vec{x},\vec{x}')}{\partial n}\right)d\vec{S}'.$$